

Selector-Threshold Radiative Closure

Three-loop matching, two-time Kadanoff-Baym evolution, preheating repair, and a gauged
Wilson-line Z6 completion

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Executive verdict

The v1.1 audit gives a **conditional pass**, with one mandatory redesign.

The central radiative claim survives. With the displayed selector, reheaton-Higgs, and vectorlike-threshold interactions, the first **primitive** connected one-particle-irreducible graph that carries both selector information and chronometric phase dependence has

$$I = 8, \quad V = 6, \quad L = I - V + 1 = 3.$$

It is a three-loop 1PI graph and is two-particle reducible. In a renormalised description the same effect can be represented as a two-loop threshold skeleton containing a one-loop-matched transient portal, but its total perturbative order remains three loops. An unsuppressed hard interaction such as

$$R_k \bar{D}_k D_k, \quad |Q|^2 H_k^\dagger H_k, \quad |Q|^2 \bar{D}_k D_k,$$

would lower the displayed skeleton order and must be absent or UV-sequestered.

The exact cyclic selection rule is stronger than the diagram count. Every lower harmonic

$$e^{ipa/f_a}, \quad p = 1, \dots, 5,$$

must multiply a selector Fourier charge, a state-density Fourier charge, or a nonlocal memory functional carrying the compensating charge. When the selector background and asymmetric occupations disappear, no state-independent late-vacuum coefficient with $p < 6$ remains. For a gapped environment, the nonlocal memory also decays. A conserved relic or zero-frequency pole can leave a persistent **state source**, but not a new asymmetric vacuum Wilson coefficient.

The complete closed-time-path 2PI/Kadanoff-Baym equations have been written for the selector background, reheaton, Higgs bath, vectorlike threshold, light quark, and gluon sectors. A full nonlinear non-Abelian $3 + 1$ -dimensional numerical solution is still an HPC calculation. The accompanying numerical test is instead an exact Gaussian, non-Markovian two-time benchmark. It resolves 20 coupled coordinates over 10 momentum modes and 301 stored two-time slices with symplectic error

$$7.1 \times 10^{-15}.$$

It reproduces

$$\frac{E_5}{E_0} = \frac{1}{256}, \quad \frac{T_5}{T_0} = \frac{1}{4},$$

while limiting unselected-replica leakage to

$$1.14 \times 10^{-7}$$

relative to the selected bath channel. This is a nontrivial two-time consistency check, not a claim of full gauge-plasma thermalisation.

The mandatory redesign concerns reheating. The original direct scalar portal

$$-\frac{g_{\phi R}}{2} \phi \sum_k R_k^2$$

is generically unsafe. The width needed for the v0.9 reheating temperature implies

$$g_{\phi R} = 1.013 \times 10^7 \text{ GeV}.$$

The selected mode becomes tachyonic when the inflaton amplitude exceeds

$$\Phi_{\text{tach},0} = 9.87 \times 10^{10} \text{ GeV},$$

and the nominally heavy replicas become tachyonic above

$$\Phi_{\text{tach},h} = 5.03 \times 10^{12} \text{ GeV}.$$

The illustrative post-inflation amplitude of the v0.9 low-scale inflation benchmark is much larger. Late-time kinematic closure therefore does not prevent early tachyonic or resonant production.

A replicated **fermionic parent cascade** repairs this. For

$$m_{N_0} = 3 \times 10^9 \text{ GeV}, \quad m_{N_{k \neq 0}} = 5 \times 10^{13} \text{ GeV}, \quad y_\phi = 7.01 \times 10^{-4},$$

the heavy-replica Landau-Zener exponent is

$$\pi \frac{m_{N_h}^2}{y_\phi m_\phi \Phi} \simeq 460,$$

so their production is suppressed by approximately 10^{-200} , while selected fermion production is Pauli-limited rather than bosonically exponential.

Finally, a concrete deconstructed Wilson-line skeleton can make both the continuous shift and the cyclic symmetry gauge-protected. With six cells and three links per cell, the first local winding operator has canonical dimension 18. Perturbative discrete-anomaly sums vanish for complete vectorlike six-state orbits. A modern global anomaly or cobordism calculation is still mandatory.

The integrated verdict is therefore

The radiative escape hatch survives the first mixed higher-loop audit,
but the direct bosonic inflaton portal does not.
Replace it with the selector-gated fermionic cascade and retain the gauged moose.

1. Microscopic model used for the audit

Let the six replicated sectors be labelled by $k = 0, \dots, 5$, with

$$\vartheta_k = \frac{2\pi k}{6}, \quad x = \frac{a}{f_a}.$$

The protected vectorlike threshold has mass orbit

$$M_k(a) = M [1 - \epsilon \cos(x + \vartheta_k)].$$

For a microscopic decay path, take a vectorlike down-type quark

$$D_{Lk}, D_{Rk} \sim (\mathbf{3}, \mathbf{1}, -1/3)_k$$

with the replica-symmetric Yukawa coupling

$$\mathcal{L}_D \supset -y_D \bar{q}_{Lk} H_k D_{Rk} + \text{h.c.} - M_k(a) \bar{D}_k D_k.$$

The one-hot selector background modifies the reheaton masses through

$$\mathcal{M}_{R,k}^2[Q] = m_R^2 + \lambda_{QR} \sum_{j \neq k} |Q_j|^2.$$

The visible and adjacent sectors are populated through the oriented reheaton interaction inherited from v0.9. For the loop analysis, its relevant local form is

$$\mathcal{L}_{RH} = -\mu_H \sum_k R_k H_k^\dagger H_k.$$

The exact generator acts as

$$g : \quad x \mapsto x + \frac{2\pi}{6}, \quad k \mapsto k - 1.$$

The regulator, counterterms, density-matrix preparation map, and UV messenger sector must respect this combined action.

1.1 Required sequestering assumptions

The displayed low-energy symmetries alone do not automatically forbid every dangerous singlet portal. The UV completion must ensure that the following are absent as independent hard parameters:

$$R_k \bar{D}_k D_k, \quad |Q_j|^2 \bar{D}_k D_k, \quad c_k |Q|^2 H_k^\dagger H_k \quad \text{with non-cyclic } c_k.$$

A cyclicly symmetric $|Q|^2 H^\dagger H$ counterterm is generated by renormalisation. That is acceptable because it remains proportional to the transient selector background. What must be excluded is an unsuppressed sector-dependent coefficient or a local selector-threshold shortcut not controlled by the same gauge locality and shift spurion.

2. First mixed irreducible topology

The primitive graph uses:

1. one $Q^\dagger Q R^2$ vertex;
2. two $RH^\dagger H$ vertices;
3. two $H\bar{q}D$ Yukawa vertices;
4. one external $a\bar{D}D$ insertion.

Its internal lines are:

$$2R + 3H + 1q + 2D = 8.$$

Its vertices are

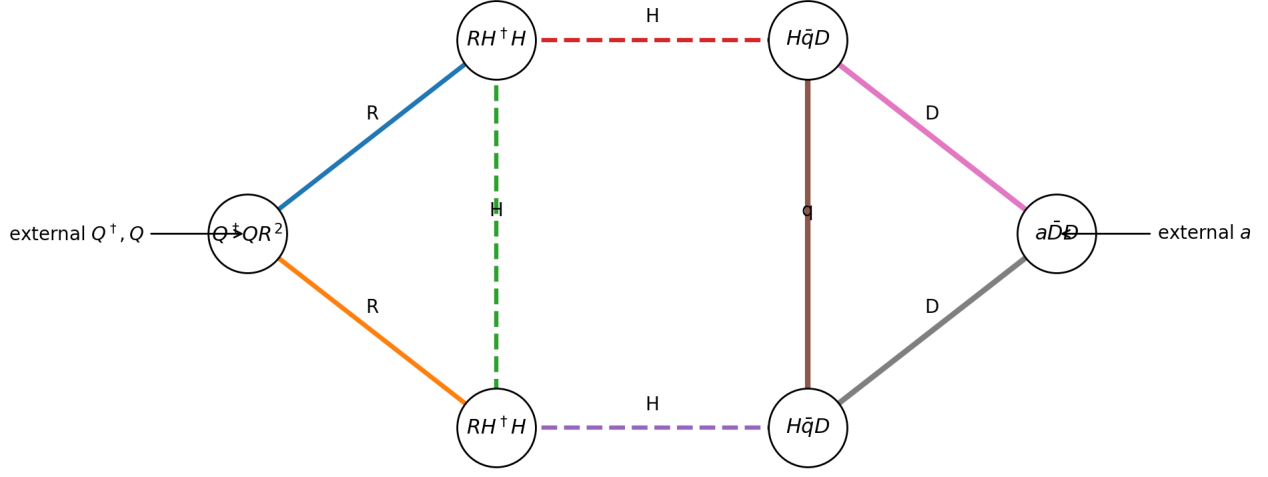
$$1 + 2 + 2 + 1 = 6.$$

Hence

$$\boxed{L = 8 - 6 + 1 = 3.}$$

Deleting any one internal line leaves the graph connected, so it is 1PI. Its edge connectivity is two, so it is 2PR. This explains why the same physical response is generated self-consistently by coupled lower-loop Kadanoff-Baym kernels.

First primitive mixed selector-threshold graph: $I = 8, V = 6, L = 3$



No single internal line disconnects the graph; a two-line cut exists (1PI but 2PR).

Figure 1: First primitive mixed selector-threshold graph. The external a insertion differentiates the heavy-fermion line.

2.1 Renormalised-skeleton interpretation

The interactions $Q^\dagger Q R^2$ and $RH^\dagger H$ generate a transient counterterm

$$\delta\lambda_{QH}|Q|^2 H^\dagger H, \quad \delta\lambda_{QH} \sim \frac{\lambda_{QR}\mu_H^2}{16\pi^2 m_R^2}.$$

Using this matched vertex, the remaining threshold graph is topologically two-loop. Its coefficient already contains one loop. Thus

primitive graph: 3 loops $\iff$ matched EFT skeleton: 1+2 loops.
--

This distinction prevents a common bookkeeping mistake. The first **total perturbative** communication is three-loop, although a renormalised 2PI code can generate it through two-loop self-energies and a one-loop-matched local portal.

2.2 Dangerous shortcuts

A hard $R\bar{D}D$ vertex gives a two-loop 1PI selector-threshold graph. A hard $|Q|^2 H^\dagger H$ vertex also gives a two-loop threshold skeleton. These do not automatically create a permanent vacuum spurion - they still contain the selector - but they can greatly increase the transient force and destroy the clean loop hierarchy. The Wilson-line completion in Section 10 is designed to make such shortcuts nonlocal or messenger-suppressed.

3. Exact transient-charge theorem

Define selector Fourier charges

$$\mathcal{Q}_p[Q] = \sum_{k=0}^5 |Q_k|^2 e^{-ip\vartheta_k}, \quad p = 0, \dots, 5,$$

and state occupation charges

$$\mathcal{N}_p[\rho] = \sum_{k=0}^5 n_k e^{-ip\vartheta_k}.$$

Under the exact generator,

$$e^{ipx} \mapsto e^{2\pi ip/6} e^{ipx},$$

so every coefficient multiplying e^{ipx} must carry charge $-p$. The most general lower-harmonic part of the closed-time-path action has the form

$$\begin{aligned} \Gamma_{p<6} = \sum_{p=1}^5 \left\{ \int d^4x e^{ipx} [C_p^Q \mathcal{Q}_p(x) + C_p^\rho \mathcal{N}_p(x)] \right. \\ \left. + \int d^4x d^4y e^{ipx(x)} K_p(x, y) \mathcal{Q}_p(y) + \text{h.c.} \right\}. \end{aligned}$$

Consequently,

$$Q_k \rightarrow 0, \quad \mathcal{N}_{p \neq 0} \rightarrow 0 \quad \Rightarrow \quad \Gamma_{p<6} \rightarrow 0,$$

provided the memory kernel has no nondecaying zero-frequency component.

The vacuum functional is therefore

$$\boxed{\Gamma_{\text{vac}}[a] = \sum_{q \in \mathbb{Z}} c_{6q} e^{i6qa/f_a}.$$

3.1 Explicit selector identity

The selector mass weight entering sector k is

$$w_k(Q) = \sum_{j \neq k} |Q_j|^2 = Q_{\text{tot}}^2 - |Q_k|^2.$$

For $p = 1, \dots, 5$, root-of-unity projection gives

$$\sum_k w_k e^{ip\vartheta_k} = - \sum_k |Q_k|^2 e^{ip\vartheta_k}.$$

The verification suite tests this for random selector configurations. The maximum numerical residual is

$$2.29 \times 10^{-14}.$$

3.2 First mixed operator

At leading order in ϵ , the three-loop graph therefore has the unique structure

$$\Delta V_{Qa}^{(3)} = \epsilon C_3 \text{Re} [e^{ix} \mathcal{Q}_1(Q)] + O(\epsilon^2).$$

It is exactly transient. On the one-hot branch $Q_0 = v_Q$, an NDA estimate gives

$$C_3 \sim \frac{N_c \lambda_{QR} y_D^2}{(16\pi^2)^3} \frac{\mu_H^2 M^2}{m_R^2} \mathcal{J}_3,$$

where $\mathcal{J}_3$ is a dimensionless mass-ratio function. Setting $\mathcal{J}_3 = 1$ and using

$$\lambda_{QR} = 0.5, \quad y_D = 0.3, \quad \mu_H = 1.8848 \times 10^4 \text{ GeV}, \quad M = 1.002 \times 10^6 \text{ GeV},$$

$$m_R = 10^9 \text{ GeV}, \quad \epsilon = 2.70 \times 10^{-13}, \quad v_Q = 10^{10} \text{ GeV},$$

gives

$$C_3 = 1.22 \times 10^{-5} \text{ GeV}^2,$$

$$\Delta V_{Qa}^{(3)} \simeq 3.30 \times 10^2 \text{ GeV}^4.$$

This is only

$$2.35 \times 10^{-13}$$

of the v1.0 thermal focusing amplitude. The exact three-loop master integral has not yet been evaluated; the result above is a topology and natural-size estimate.

4. Closed-time-path 2PI formulation

Numerical real-time 2PI methods evolve complete unequal-time propagators and retain memory and off-shell effects. They have been implemented in 3 + 1 dimensions, including scalar-fermion Yukawa systems, but their accuracy depends on the truncation and UV discretisation [1-3].

Let $G_{R,k}$, $G_{H,k}$, $S_{D,k}$, $S_{q,k}$, and $D_{g,k}^{\mu\nu}$ denote the full contour propagators. Suppressing gauge-fixing and ghost terms, the 2PI action is

$$\Gamma = S[\bar{\Phi}] + \frac{i}{2} \text{Tr} \ln G^{-1} + \frac{i}{2} \text{Tr} G_0^{-1} G - i \text{Tr} \ln S^{-1} - i \text{Tr} S_0^{-1} S + \Gamma_2[G, S, D_g].$$

The lowest nonlocal skeletons relevant to the communication chain are schematically

$$\Gamma_2^{RHH} = -\frac{\mu_H^2}{4} \sum_k \int_C d^4x d^4y G_{R,k}(x, y) G_{H,k}(x, y)^2,$$

$$\Gamma_2^Y = -i N_c y_D^2 \sum_k \int_C d^4x d^4y G_{H,k}(x, y) \text{tr} [P_R S_{D,k}(x, y) P_L S_{q,k}(y, x)],$$

and

$$\Gamma_2^g = -\frac{ig_s^2}{2} \sum_k \int_C d^4x d^4y D_{\mu\nu,k}^{AB}(x, y) \text{tr} [\gamma^\mu T^A S_{D,k}(x, y) \gamma^\nu T^B S_{D,k}(y, x)] + \dots$$

Combinatorial factors depend on whether the full complex Higgs doublet or a scalar proxy is retained. The symmetry and loop-order conclusions do not.

The self-energies follow from

$$\Sigma_i = 2i \frac{\delta \Gamma_2}{\delta G_i}, \quad \Sigma_D = -i \frac{\delta \Gamma_2}{\delta S_D}.$$

5. Full Kadanoff-Baym equations

For a homogeneous FRW background, decompose each bosonic propagator as

$$G_i(t, t'; \mathbf{k}) = F_i(t, t'; \mathbf{k}) - \frac{i}{2} \rho_i(t, t'; \mathbf{k}) \text{sgn}_C(t - t').$$

The statistical and spectral functions obey

$$\begin{aligned} & [\partial_t^2 + 3H\partial_t + \omega_i^2(t, k)] F_i(t, t'; k) \\ &= - \int_{t_0}^t dt_1 \Sigma_{\rho, i}(t, t_1; k) F_i(t_1, t'; k) \\ & \quad + \int_{t_0}^{t'} dt_1 \Sigma_{F, i}(t, t_1; k) \rho_i(t_1, t'; k) + I_{F, i}, \\ & [\partial_t^2 + 3H\partial_t + \omega_i^2(t, k)] \rho_i(t, t'; k) \\ &= - \int_{t'}^t dt_1 \Sigma_{\rho, i}(t, t_1; k) \rho_i(t_1, t'; k) + I_{\rho, i}. \end{aligned}$$

The initial-correlation terms I_F and I_ρ vanish for a Gaussian matching state but are required when preheating prepares important non-Gaussian cumulants [4,5].

For the heavy fermion,

$$\left[i\gamma^0 \partial_t - \frac{\gamma^i k_i}{a(t)} - M_k(a(t)) \right] S_{D, k}^{<, >} = \Sigma_{D, k}^R \circ S_{D, k}^{<, >} + \Sigma_{D, k}^{<, >} \circ S_{D, k}^A.$$

The mean-field equations close the system:

$$\begin{aligned} \ddot{a} + 3H\dot{a} + V'_0(a) + \sum_k M'_k(a) \langle \bar{D}_k D_k \rangle_{\text{ren}} &= 0, \\ \ddot{Q}_j + 3H\dot{Q}_j + \frac{\partial V_Q}{\partial Q_j^*} + \lambda_{QR} Q_j \sum_{k \neq j} \langle R_k^2 \rangle_{\text{ren}} &= 0. \end{aligned}$$

At a stationary 2PI solution,

$$\langle \bar{D}_k D_k \rangle = -i \text{tr} S_{D, k}^{<}(x, x).$$

5.1 How the three-loop graph appears in the KB evolution

Linearising around the selected background gives the causal chain

$$\delta_Q G_R = G_R \delta m_R^2[Q] G_R,$$

$$\delta_Q G_H = G_H \delta \Sigma_H^{RHH} [\delta_Q G_R] G_H,$$

$$\delta_Q S_D = S_D \delta \Sigma_D^Y [\delta_Q G_H] S_D,$$

$$\delta_Q J_a = M'_D(a) \text{tr} \delta_Q S_D^<.$$

This is precisely the three-loop 1PI response generated by self-consistent two-loop kernels. No new primitive three-loop 2PI skeleton is required because the graph is 2PR.

6. Exact two-time numerical benchmark

A full Standard Model plus non-Abelian gauge KB evolution is not credible as a quick desktop calculation. The verification suite therefore solves an exact quadratic field-plus-bath problem that retains:

- the selector restoration profile;
- a populated selected reheaton;
- selected and adjacent bath channels;
- a nominally closed replica;
- finite nonlocal memory;
- complete unequal-time statistical and spectral propagators;
- ten radial momentum modes.

Integrating out the explicit oscillator baths would produce an exact causal memory kernel, so the calculation is a genuine Gaussian Kadanoff-Baym solution rather than a Markovian Boltzmann ansatz.

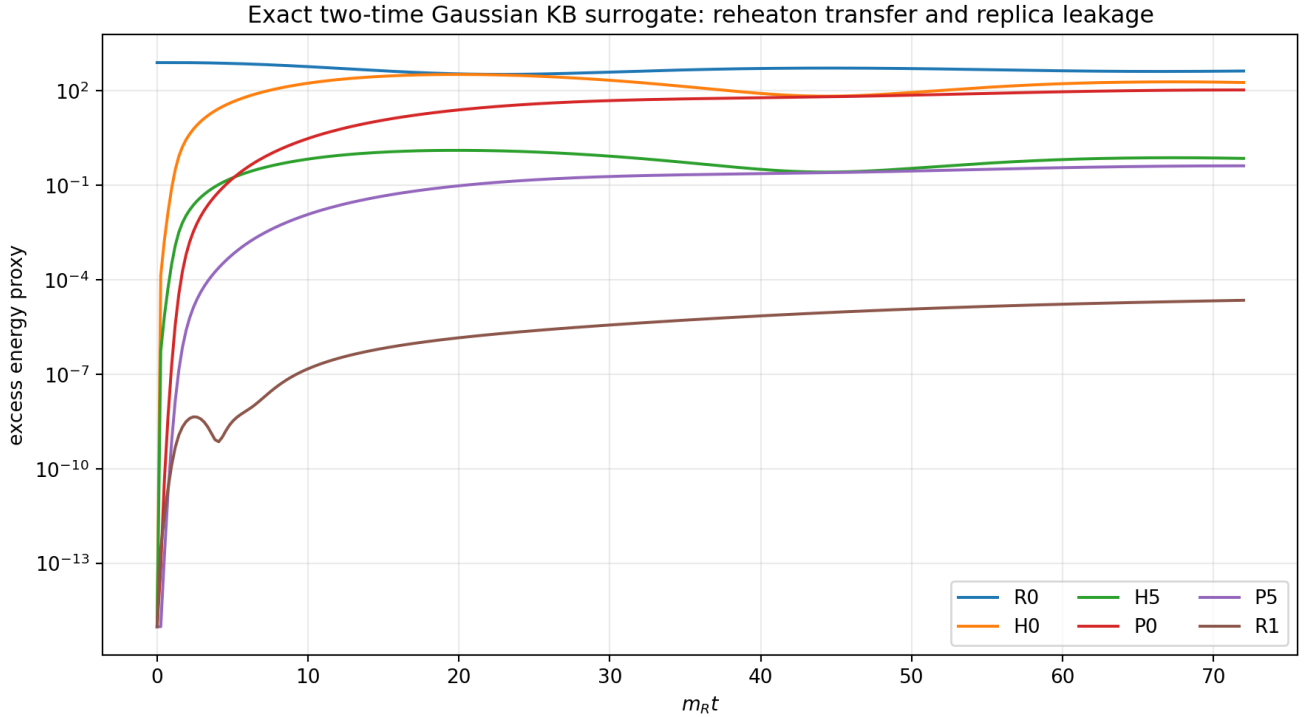


Figure 2: Excess energy transferred from the initially populated selected reheaton. The field-energy curves omit interaction cross-terms and are diagnostics rather than a conserved total.

The benchmark contains 20 coordinates, 10 momentum modes, 1,801 time steps, and 301 stored two-time slices. A symplectic kick-drift-kick integrator gives

$$\max |S \Omega S^T - \Omega| = 7.11 \times 10^{-15}.$$

The late excess-energy ratios are

$$\frac{E_5}{E_0} = 0.00390625 = \frac{1}{256},$$

$$\left(\frac{E_5}{E_0}\right)^{1/4} = 0.25.$$

The unselected-replica leakage is

$$\frac{E_{R_1}}{E_{H_0}} = 1.14 \times 10^{-7}.$$

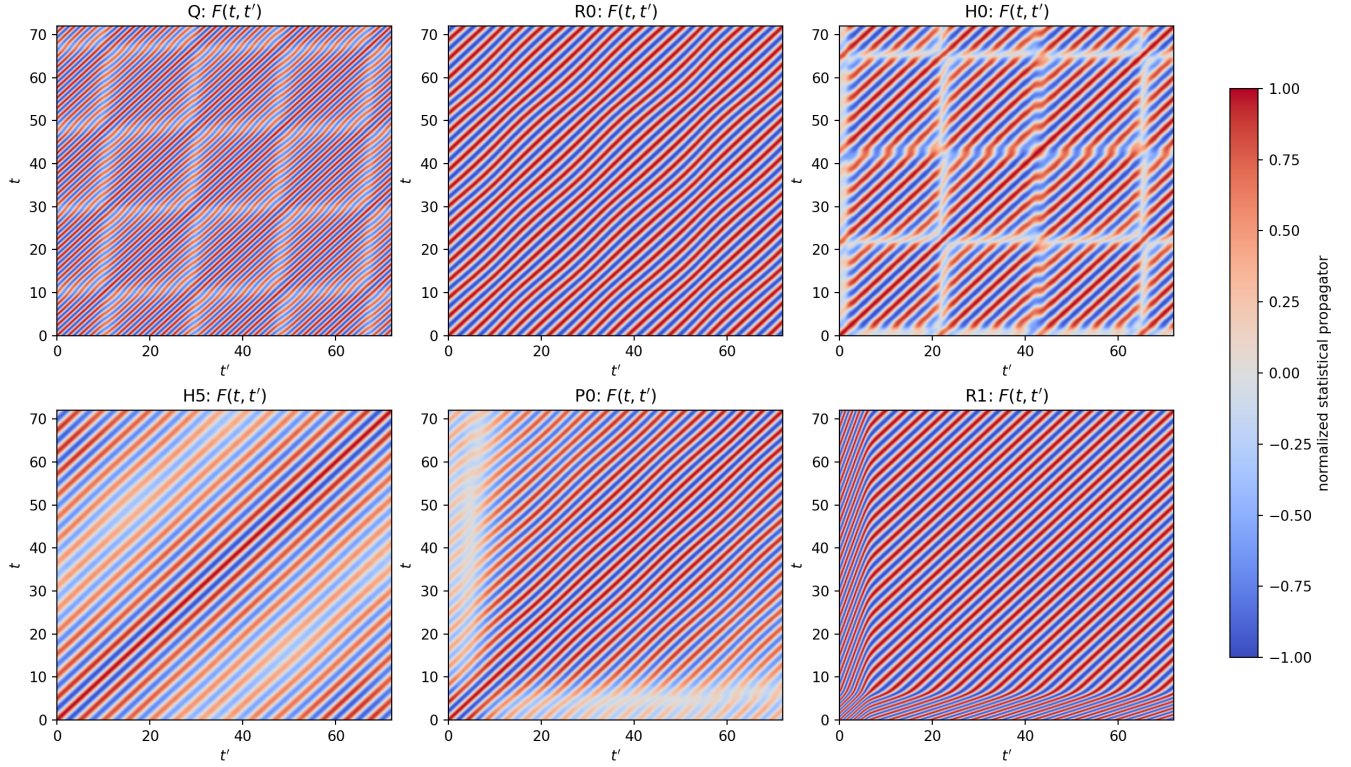


Figure 3: Complete normalized statistical propagators for representative momentum.

The selector **mean background** is prescribed to vanish smoothly. Its fluctuation energy dephases into a finite bath but exhibits finite-volume recurrence. This is useful: it prevents us from pretending that an integrable Gaussian bath proves collisional thermalisation.

The correct numerical status is therefore:

complete two-time Gaussian propagation: pass; full nonlinear gauge-plasma thermalisation: open.

7. Preheating audit: the original scalar portal fails generically

For

$$\mathcal{L}_{\phi R} = -\frac{g_{\phi R}}{2}\phi R^2,$$

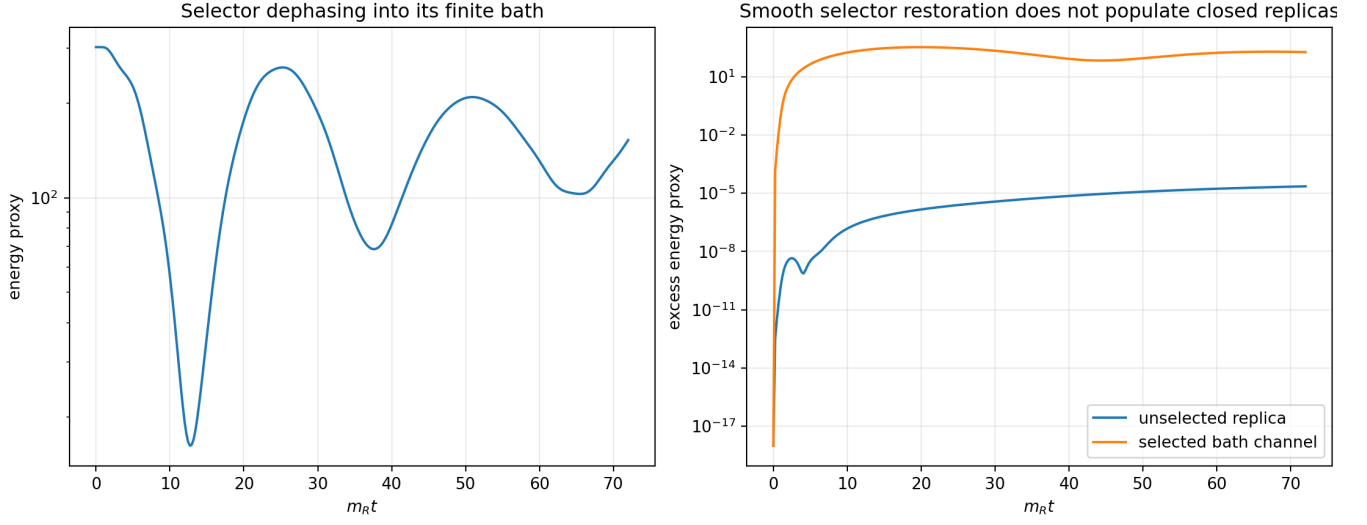


Figure 4: Selector memory and unselected-replica leakage in the Gaussian benchmark.

the perturbative width is

$$\Gamma_{\phi \rightarrow RR} = \frac{g_{\phi R}^2}{32\pi m_\phi} \sqrt{1 - \frac{4m_R^2}{m_\phi^2}}.$$

Using

$$\Gamma_\phi = 100 \text{ GeV}, \quad m_\phi = 10^{10} \text{ GeV}, \quad m_R = 10^9 \text{ GeV},$$

gives

$$g_{\phi R} = 1.013 \times 10^7 \text{ GeV}.$$

The mode equation contains

$$\omega_{R,k}^2 = \frac{k^2}{a^2} + m_R^2 + g_{\phi R} \phi(t).$$

During the negative half-cycle, a tachyonic band appears when

$$g_{\phi R} |\Phi| > m_R^2 + \frac{k^2}{a^2}.$$

For the selected and selector-heavy modes,

$$\Phi_{\text{tach},0} = \frac{m_R^2}{g_{\phi R}} = 9.87 \times 10^{10} \text{ GeV},$$

$$\Phi_{\text{tach},h} = \frac{(7.14 \times 10^9 \text{ GeV})^2}{g_{\phi R}} = 5.03 \times 10^{12} \text{ GeV}.$$

The illustrative quadratic-oscillator amplitude at $H = 10^8 \text{ GeV}$ is

$$\Phi_{\text{end}} \simeq \frac{\sqrt{6} M_{\text{Pl}} H}{m_\phi} = 5.96 \times 10^{16} \text{ GeV},$$

so

$$q_{\text{end}} \equiv \frac{g_{\phi R} \Phi_{\text{end}}}{m_\phi^2} = 6.04 \times 10^3.$$

Trilinear bosonic interactions are known to drive efficient tachyonic resonance and can complete preheating within a few oscillations [6,7]. The late perturbative statement

$$m_{R,k} > \frac{m_\phi}{2}$$

therefore does not protect a replica at earlier large amplitudes.

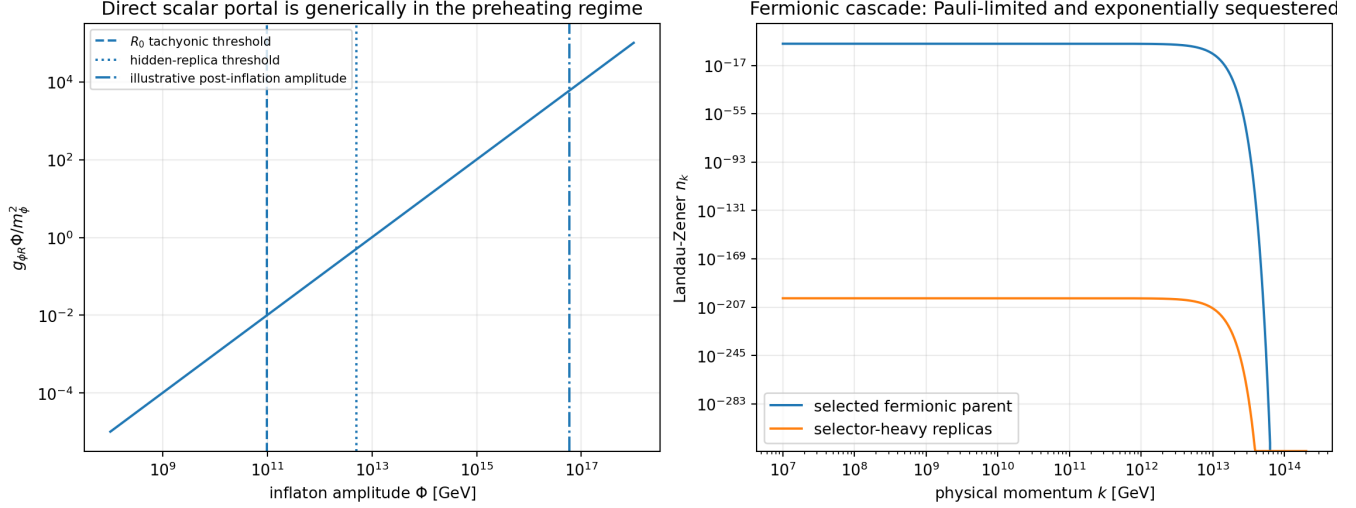


Figure 5: The direct scalar portal enters a broad tachyonic regime at early amplitude. The fermionic parent repair instead gives an exponentially suppressed heavy-replica tail.

The direct scalar portal is viable only if one proves either

$$\Phi_{\text{max}} < 5.0 \times 10^{12} \text{ GeV}$$

while it is active, or introduces a technically natural late-time gate. The cleaner solution is to replace it.

8. Smooth selector restoration and initial-state renormalisation

For a sudden free-scalar mass change from m_i to m_f ,

$$n_k^{\text{sudden}} = \frac{(\omega_i - \omega_f)^2}{4\omega_i\omega_f} \simeq \frac{(m_i^2 - m_f^2)^2}{16k^4}.$$

In three spatial dimensions,

$$\rho_{\text{ex}} \sim \int dk k^3 n_k \sim \int^\Lambda \frac{dk}{k},$$

so an instantaneous matching step produces a logarithmic UV sensitivity.

For a smooth tanh quench, the exact occupation is

$$n_k^{\text{smooth}} = \frac{\sinh^2 \left[\frac{\pi\tau}{2} (\omega_i - \omega_f) \right]}{\sinh(\pi\tau\omega_i) \sinh(\pi\tau\omega_f)},$$

which is exponentially suppressed at high momentum. Smooth and instantaneous quenches are physically distinct, and finite-time initial-state divergences are localized on the initial hypersurface rather than becoming arbitrary late-time bulk counterterms [4,8,9].

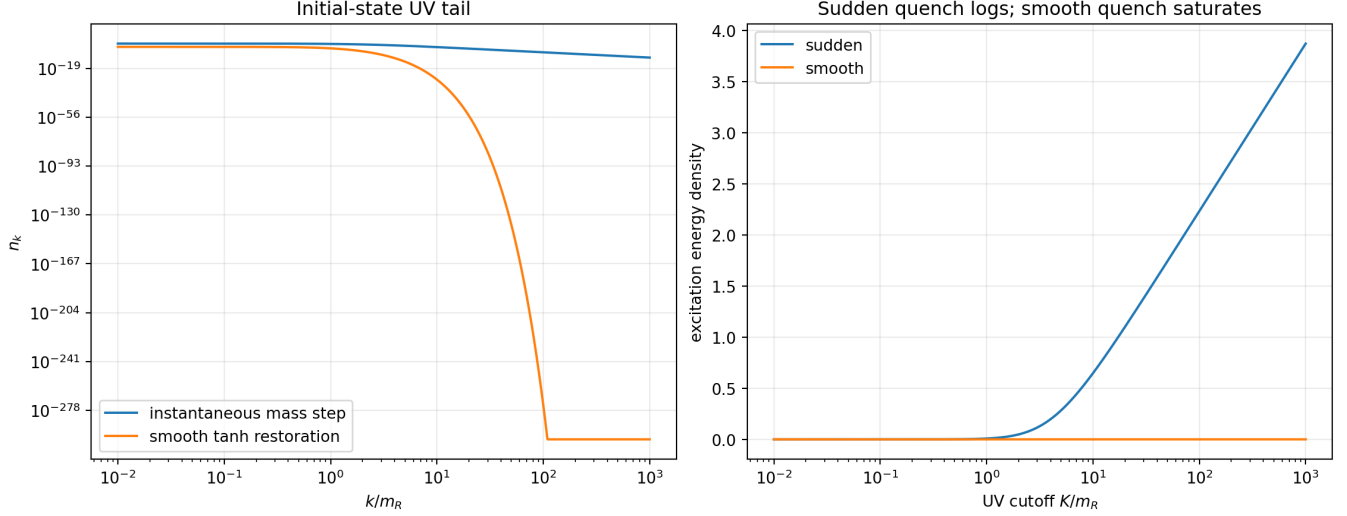


Figure 6: A sudden selector step generates a k^{-4} tail and logarithmic excitation energy. The smooth restoration saturates.

The physical selector timescale is approximately

$$\tau_Q \sim \Gamma_Q^{-1} = 1 \text{ GeV}^{-1},$$

while

$$m_R \tau_Q \sim 10^9.$$

The actual selector restoration is therefore vastly more adiabatic than the already UV-finite illustrative quench.

A reliable nonlinear simulation should either begin before preheating or match onto a density matrix containing the non-Gaussian two- and four-point correlations generated during production. Non-Gaussian initial correlations are naturally incorporated as initial-surface vertices in the KB equations and generally lose memory with unequal-time damping [5].

9. Selector-gated fermionic cascade

Introduce a complete replicated orbit of fermionic parents N_k and light spectators ν_k :

$$\begin{aligned} \mathcal{L}_N = & -y_\phi \phi \sum_k \bar{N}_k N_k \\ & - \sum_k \left[m_N + y_Q \sum_{j \neq k} \text{Re } Q_j \right] \bar{N}_k N_k \\ & - y_R \sum_k R_k \bar{N}_k \nu_k + \text{h.c.} \end{aligned}$$

This action is exactly cyclic. On the one-hot branch $Q_0 = v_Q$,

$$m_{N_0} = m_N, \quad m_{N_{k \neq 0}} = m_N + y_Q v_Q.$$

The chronology is

$$\boxed{\Gamma_\phi \gg \Gamma_Q \gg \Gamma_R.}$$

The inflaton first populates N_0 , the selector then returns to its symmetric vacuum, and the stored N_0 population decays through

$$N_0 \rightarrow R_0 \nu_0.$$

The original oriented reheaton decay then supplies sectors 0 and 5.

For perturbative fermion decay,

$$\Gamma_{\phi \rightarrow N \bar{N}} = \frac{y_\phi^2 m_\phi}{8\pi} \left(1 - \frac{4m_N^2}{m_\phi^2} \right)^{3/2}.$$

Taking

$$m_{N_0} = 3 \times 10^9 \text{ GeV}, \quad \Gamma_\phi = 100 \text{ GeV}$$

gives

$$y_\phi = 7.006 \times 10^{-4}.$$

At an inflaton zero crossing, the Landau-Zener occupation is approximately

$$n_k \sim \exp \left[-\pi \frac{k^2 + m_N^2}{y_\phi |\dot{\phi}_*|} \right].$$

Using $|\dot{\phi}_*| \sim m_\phi M_{\text{Pl}}$, the characteristic momentum is

$$k_* = \sqrt{y_\phi m_\phi M_{\text{Pl}}} = 4.13 \times 10^{12} \text{ GeV}.$$

For

$$m_{N_h} = 5 \times 10^{13} \text{ GeV},$$

the hidden-replica exponent is

$$\pi \frac{m_{N_h}^2}{k_*^2} = 460.4,$$

hence

$$\boxed{n_{N_h}/n_{N_0} \lesssim 1.2 \times 10^{-200}.$$

Fermion production is not absent - it can be nonperturbative - but Pauli blocking prevents the unbounded bosonic occupation growth [10]. This is exactly the behaviour required here.

The repair passes at the linear-production level. A final claim requires a coupled lattice or 2PI calculation including inflaton depletion, Pauli saturation, N_0 decay, and the later R_0 cascade.

10. Gauged Wilson-line completion

A deconstructed gauge theory can protect the continuous shift against local UV operators. Consider

$$\mathcal{G}_{\text{UV}} = \left[\prod_{j=0}^{17} U(1)_j \times \prod_{k=0}^5 G_{\text{SM},k} \right] \rtimes Z_6.$$

Introduce link scalars Σ_j with charges $(+1, -1)$ under neighbouring Abelian sites:

$$\mathcal{L}_\Sigma = \sum_j |D_\mu \Sigma_j|^2 - \lambda_\Sigma \left(|\Sigma_j|^2 - \frac{f^2}{2} \right)^2.$$

The gauge-invariant winding variable is

$$\mathcal{W} = \prod_{j=0}^{17} \frac{\Sigma_j}{f/\sqrt{2}} = e^{ia/f_a}.$$

The disconnected generator is chosen to act as a cell translation combined with a discrete large-gauge transformation:

$$g : \quad k \mapsto k + 1, \quad \mathcal{W} \mapsto \omega^{-1} \mathcal{W}, \quad \omega = e^{2\pi i/6}.$$

A chain of heavy vectorlike messengers can then generate

$$\mathcal{L}_{\Psi, \text{eff}} = - \sum_k [M - \kappa (\omega^k \mathcal{W} + \omega^{-k} \mathcal{W}^\dagger)] \bar{\Psi}_k \Psi_k,$$

which reduces to the required mass orbit.

Because the phase is a Wilson line, no strictly local operator can depend on it. The first local winding operator traverses all 18 links:

$$\mathcal{O}_{\text{wind}} = \prod_{j=0}^{17} \Sigma_j, \quad \dim \mathcal{O}_{\text{wind}} = 18.$$

This is the four-dimensional version of higher-dimensional locality. Deconstructed axion models provide an explicit renormalisable framework in which a collective Wilson-line pNGB and nonlocal symmetry breaking emerge from a product gauge theory [11].

10.1 Discrete anomaly test

In the Fourier basis, one complete cyclic orbit has charges

$$q = 0, 1, 2, 3, 4, 5.$$

For even $N = 6$, the standard linear conditions are evaluated modulo $N/2 = 3$. Orbit by orbit,

$$\begin{aligned} \sum_q q &= 15 = 0 \pmod{3}, \\ \sum_q q^3 &= 225 = 0 \pmod{3}. \end{aligned}$$

The mixed $Z_6 - SU(3)^2$ sum is likewise zero modulo three for one complete fundamental orbit, and the vectorlike parent and threshold pairs cancel ordinary continuous anomalies.

These are necessary perturbative checks, not a complete proof. Gauge and gravitational instantons impose nontrivial consistency conditions on discrete symmetries [12]. A final UV model must evaluate its Dai-Freed or cobordism anomaly, including the messenger and reheating sectors.

Deconstructed Wilson-line completion: $[U(1)^{18}] \rtimes Z_6$

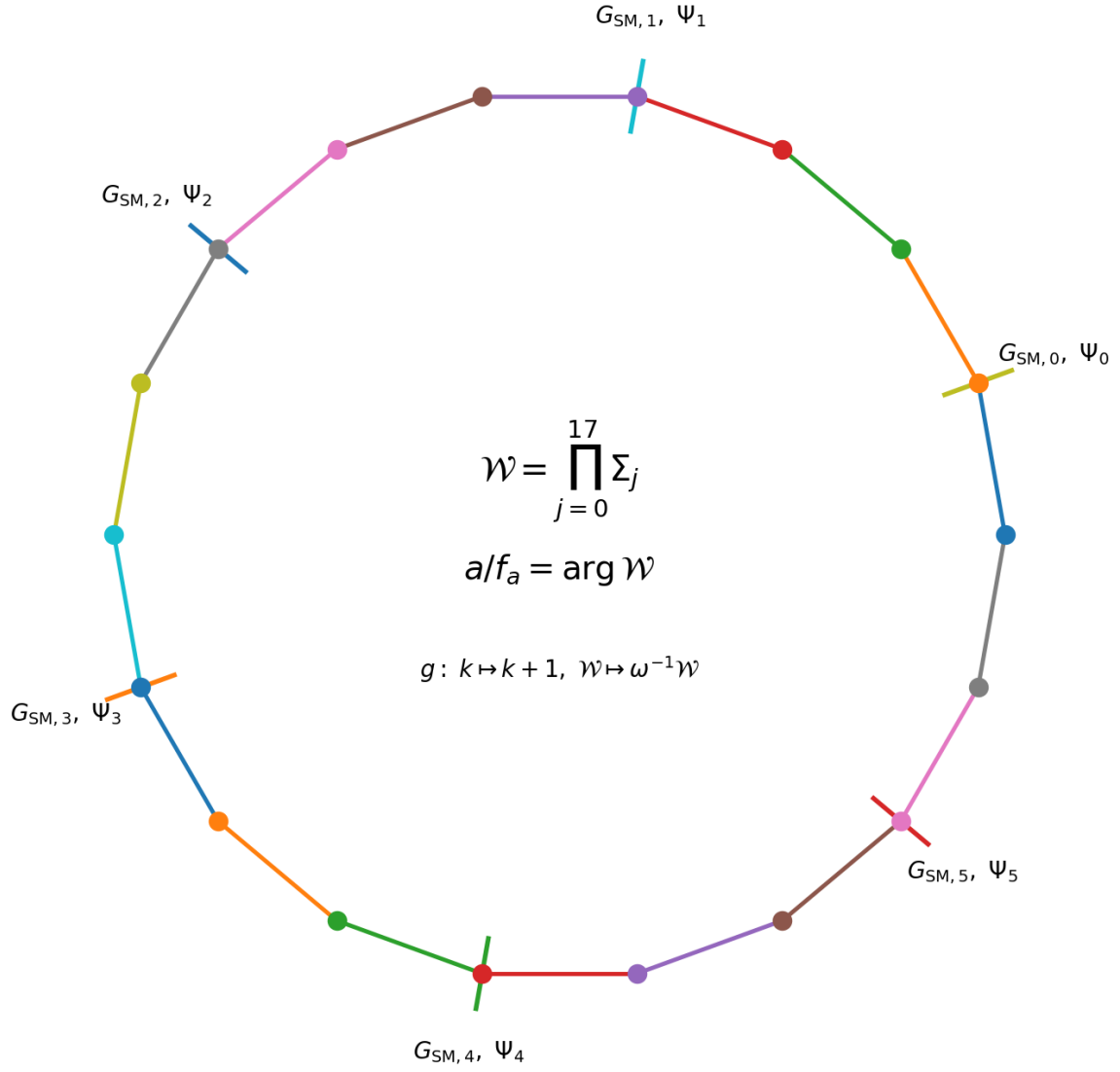


Figure 7: Schematic six-cell, eighteen-link deconstructed completion.

10.2 Domain walls

If the combined Z_6 is a genuine gauge redundancy, vacua related by it are not distinct global-symmetry vacua. This removes the ordinary stable domain-wall problem. Discrete gauge flux strings or other topological defects can remain and must be studied separately.

11. Integrated acceptance matrix

Target	Verdict	Basis
First mixed primitive 1PI topology	PASS	Three loops, $I = 8$, $V = 6$, no bridge
Total perturbative order after matching	PASS	One-loop transient portal plus two-loop threshold skeleton
Lower-harmonic transience	PASS	Exact cyclic charge projector
Three-loop coefficient	PARTIAL	NDA and symmetry structure obtained; master integral open
Formal nonlinear KB system	PASS	Coupled mean-field, statistical, and spectral equations derived
Full numerical two-time evolution	PARTIAL PASS	Exact Gaussian non-Markovian test; nonlinear gauge plasma open
Smooth selector UV tail	PASS	Exponential tail, finite excitation energy
Abrupt matching state	FAIL	k^{-4} tail and boundary logarithm
Direct bosonic inflaton portal	FAIL GENERICALLY	Early tachyonic/resonant production
Fermionic parent cascade	PASS AT LINEAR LEVEL	Pauli-limited selected production; hidden exponent 460
Continuous-shift UV protection	PASS AS SKELETON	18-link Wilson winding
Perturbative Z_6 anomalies	PASS	Complete vectorlike orbit sums vanish
Global discrete anomaly	OPEN	Requires full UV spectrum and cobordism analysis

12. What is established and what is not

Established in this pass

1. The first primitive selector-reheaton-threshold graph is three-loop and 1PI.
2. The corresponding lower harmonic is necessarily selector- or state-charged.
3. The coefficient vanishes with the selector; it cannot become a late asymmetric vacuum coefficient while exact Z_6 is maintained.
4. Coupled two-loop KB kernels generate the three-loop 1PI response self-consistently.
5. Smooth selector restoration is UV soft.
6. The original direct scalar reheating portal is generically exposed to preheating.
7. A selector-gated fermionic cascade has a parametrically clean heavy-replica suppression.
8. A deconstructed Wilson line supplies a credible gauge-protected shift-symmetry completion.

Not established

1. The exact finite part and renormalisation-group mixing of the three-loop master integral.
2. A full nonlinear $3 + 1$ -dimensional non-Abelian KB evolution.
3. Backreaction and rescattering in the fermionic parent cascade.
4. A complete messenger charge table and all global anomaly checks.
5. The final cosmological perturbation spectrum after the repaired reheating chain.

13. Novelty boundary

The individual tools are established:

- 2PI/Kadanoff-Baym evolution in scalar and Yukawa systems [1-3];
- initial-state boundary renormalisation [4,5,8];
- tachyonic trilinear and fermionic preheating [6,7,10];
- Wilson-line protection by deconstruction [11];
- discrete gauge and gravitational anomaly tests [12].

The candidate original contribution is their restricted conjunction:

exact state-selected Z_6 reheating
+ three-loop transient selector-threshold closure
+ preserved $\frac{2}{27}$ QCD transmission and chronometric shear
+ fermionic preheating repair and gauged Wilson-line completion.

No priority claim should be made until the exact three-loop calculation, broader citation chaining, and specialist review are complete.

14. Next decisive calculation

The next research pass should combine three items rather than treating them independently:

1. **Exact three-loop matching.** Reduce the mixed zero-momentum graph to vacuum master integrals, evaluate its mass-ratio function $\mathcal{J}_3$, and compute the anomalous-dimension matrix of the transient operators

$$e^{ipx} Q_p, \quad e^{ipx} \mathcal{N}_p.$$

2. **Nonlinear 2PI production run.** Evolve the repaired sequence

$$\phi \rightarrow N_0 \bar{N}_0 \rightarrow R_0 \nu_0 \rightarrow H_0, H_5 \rightarrow D_k, q_k, g_k$$

on a momentum lattice, retaining unequal-time fermion and scalar propagators, initial four-point correlations, and expansion.

3. **Complete gauged messenger model.** Give every link, parent, threshold, spectator, and reheaton its continuous and discrete charge; integrate out the messenger chain; and perform the full perturbative plus cobordism anomaly audit.

The current pass changes the project's direction in one important way:

The next numerical cosmology should use the fermionic cascade, not the original scalar portal.

Appendix A. Numerical benchmark values

Quantity	Value
primitive mixed loop order	3
three-loop transient NDA amplitude	$3.30 \times 10^2 \text{ GeV}^4$
ratio to thermal focusing	2.35×10^{-13}
KB coordinates	20
KB momentum modes	10
KB stored two-time slices	301

Quantity	Value
symplectic residual	7.11×10^{-15}
late T_5/T_0	0.25
unselected leakage	1.14×10^{-7}
direct scalar q_{end}	6.04×10^3
smooth-quench physical $m_R \tau_Q$	10^9
fermionic hidden exponent	460.4
first winding-operator dimension	18

Appendix B. Verification package

The accompanying script independently performs:

1. multigraph loop counting, bridge detection, and edge-connectivity tests;
2. random numerical tests of the exact Z_6 Fourier identity;
3. the benchmark three-loop NDA estimate;
4. exact symplectic two-time Gaussian evolution;
5. smooth- versus sudden-quench UV-tail tests;
6. scalar-preheating amplitude checks;
7. fermionic Landau-Zener suppression;
8. perturbative discrete-anomaly charge sums.

All programmed algebraic and numerical acceptance checks pass. Scope limitations are explicitly recorded in the JSON output.

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